Sprawozdanie LAB 5

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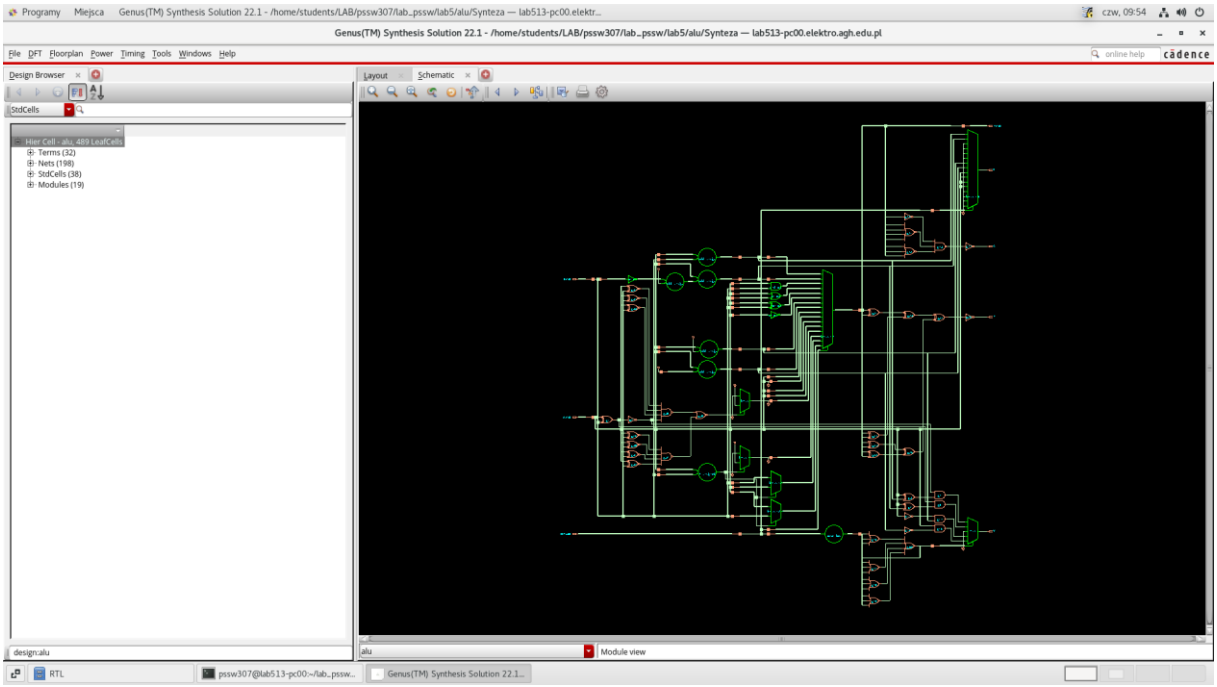
Pssw307

412065

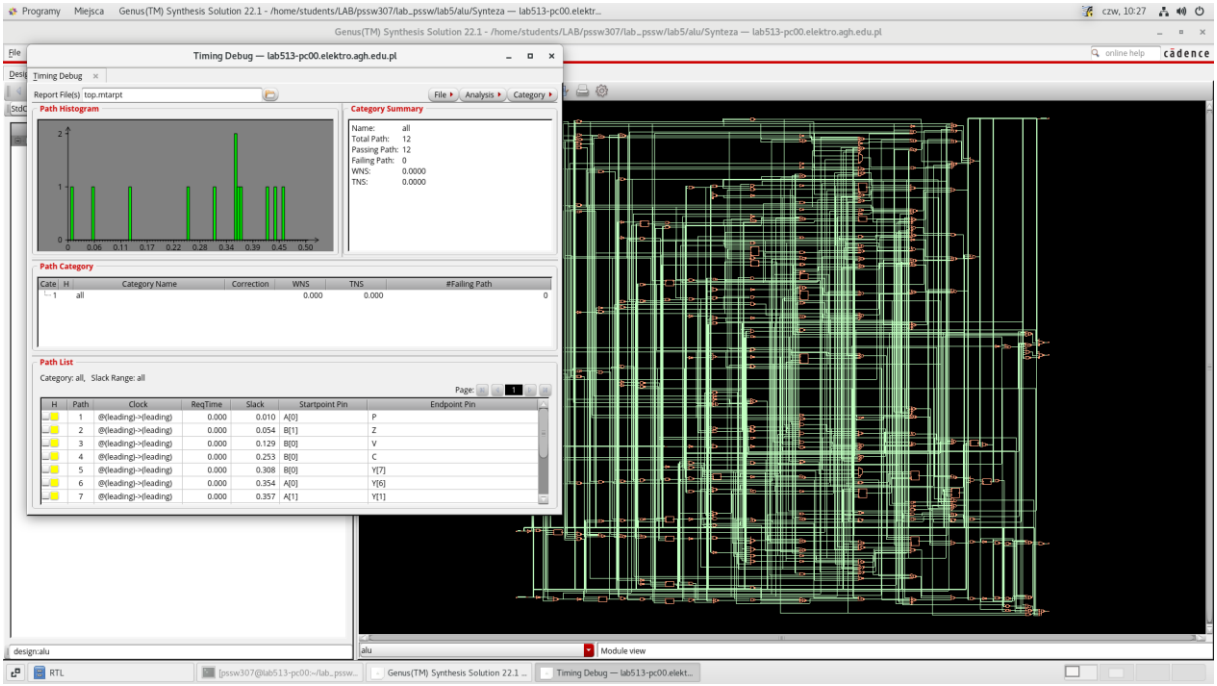
1. ALU

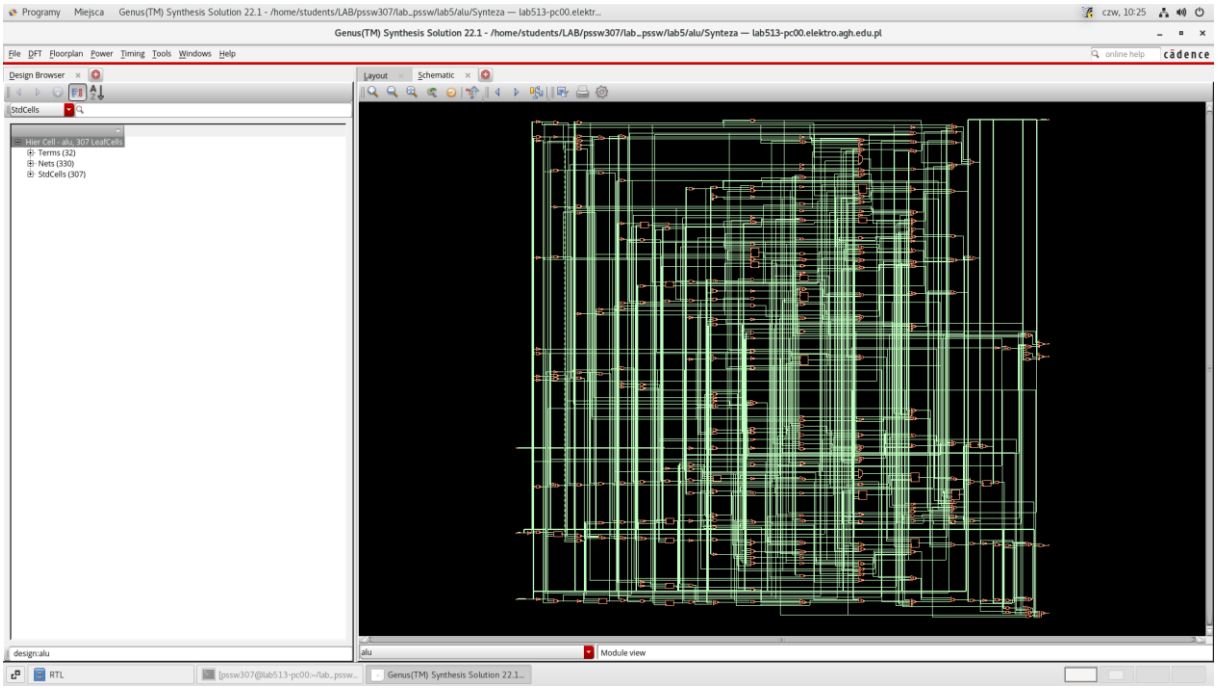
Max opóźnienie 1ns z zapasem 10ps.

1.1. Przed syntezą



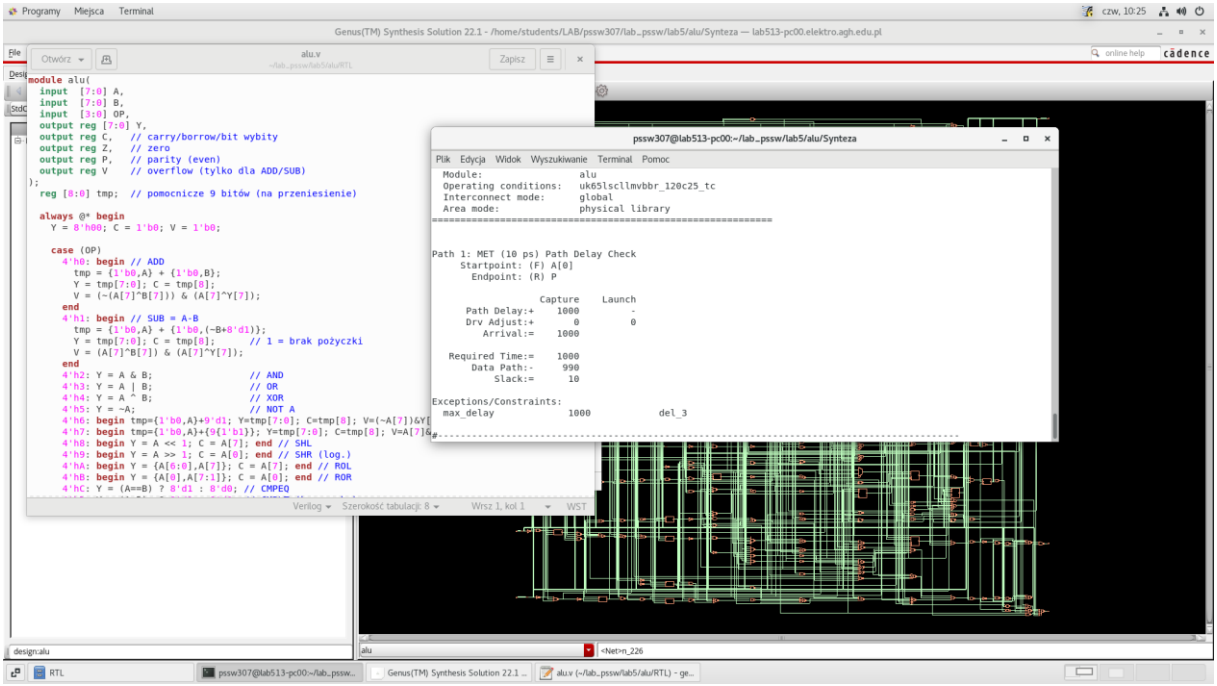
1.2. Po poprawnej optymalizacji i syntezie

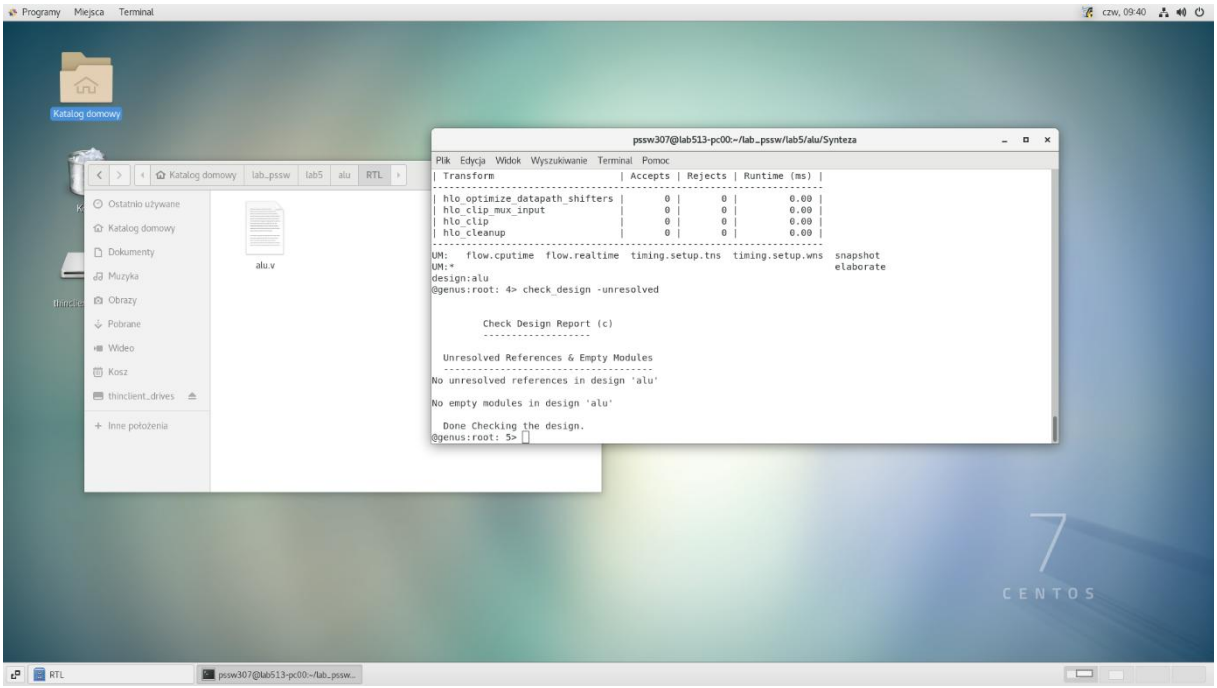




1.3. Wynik testów weryfikacyjnych

```
================================================================================
Mapped points: SYSTEM class
--------------------------------------------------------------------------------
Mapped points     PI     PO        Total
--------------------------------------------------------------------------------
Golden            20     12        32
--------------------------------------------------------------------------------
Revised           20     12        32
================================================================================
// Command: add compared point -all
// 12 compared points added to compare list
// Command: compare
================================================================================
Compared points      PO        Total
--------------------------------------------------------------------------------
Equivalent           12        12
================================================================================
// Command: report verification
================================================================================
                            Verification Report
--------------------------------------------------------------------------------
Category                                                                  Count
--------------------------------------------------------------------------------
1. Non-standard modeling options used:                                      0
--------------------------------------------------------------------------------
2. Incomplete verification:                                                 0
--------------------------------------------------------------------------------
3. User modification to design:                                             0
--------------------------------------------------------------------------------
4. Conformal Constraint Designer clock domain crossing checks recommended:  0
--------------------------------------------------------------------------------
5. Design ambiguity:                                                        0
--------------------------------------------------------------------------------
6. Compare Results:                                                      PASS
================================================================================
// Command: report verification > pssw307_alu_report.txt
// Command: exit
```

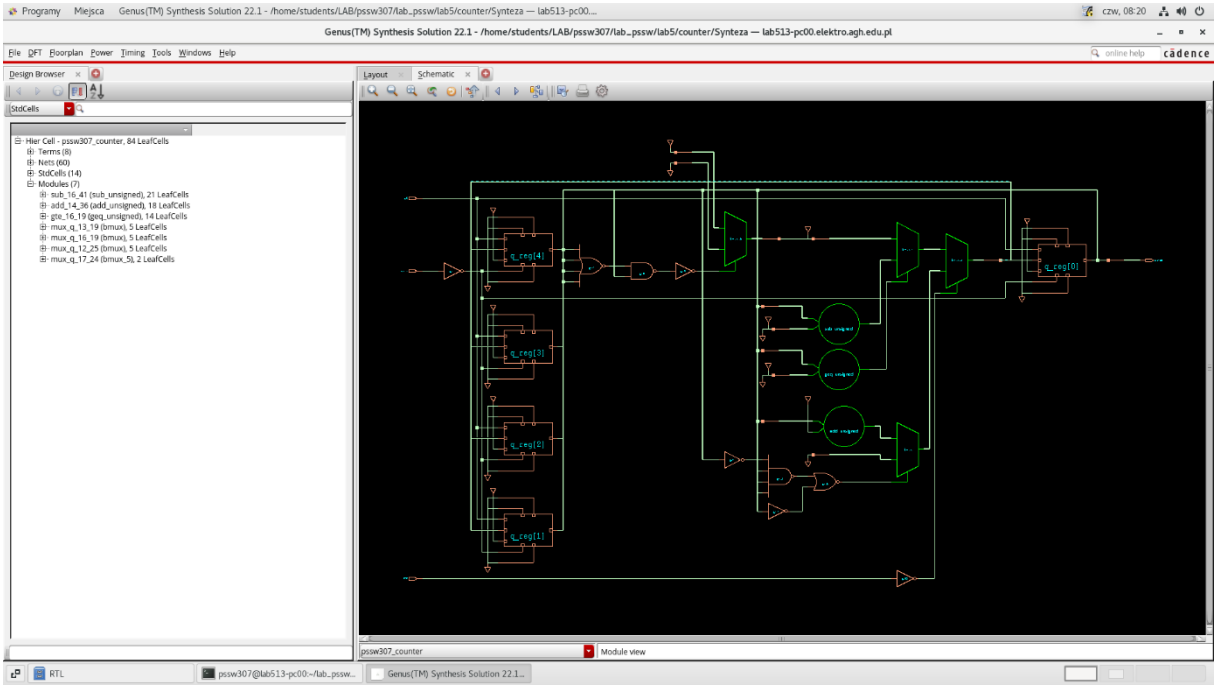

Programy Miejsca Terminal
czw, 10:25
Genus(TM) Synthesis Solution 22.1 - /home/students/LAB/pssw307/lab_pssw/lab5/alu/Synteza — lab513-pc00.elektro.agh.edu.pl
alu.v
Otwórz
Zapisz
module alu(
input [7:0] A,
input [7:0] B,
input [3:0] OP,
output reg [7:0] Y,
output reg C, // carry/borrow/bit wybity
output reg Z, // zero
output reg P, // parity (even)
output reg V // overflow (tylko dla ADD/SUB)
);
reg [8:0] tmp; // pomocnicze 9 bitów (na przeniesienie)
always @* begin
Y = 8'h00; C = 1'b0; V = 1'b0;
case (OP)
4'h0: begin // ADD
tmp = {1'b0,A} + {1'b0,B};
Y = tmp[7:0]; C = tmp[8];
V = (~(A[7]^B[7])) & (A[7]^Y[7]);
end
4'h1: begin // SUB = A-B
tmp = {1'b0,A} + {1'b0,(~B+8'd1)};
Y = tmp[7:0]; C = tmp[8]; // 1 = brak pożyczki
V = (A[7]^B[7]) & (A[7]^Y[7]);
end
4'h2: Y = A & B; // AND
4'h3: Y = A | B; // OR
4'h4: Y = A ^ B; // XOR
4'h5: Y = ~A; // NOT A
4'h6: begin tmp={1'b0,A}+9'd1; Y=tmp[7:0]; C=tmp[8]; V=(~A[7])&Y[
4'h7: begin tmp={1'b0,A}+{9{1'b1}}; Y=tmp[7:0]; C=tmp[8]; V=A[7]&
4'h8: begin Y = A << 1; C = A[7]; end // SHL
4'h9: begin Y = A >> 1; C = A[0]; end // SHR (log.)
4'hA: begin Y = {A[6:0],A[7]}; C = A[7]; end // ROL
4'hB: begin Y = {A[0],A[7:1]}; C = A[0]; end // ROR
4'hC: Y = (A==B) ? 8'd1 : 8'd0; // CMPEQ
Verilog Szerokość tabulacji: 8 Wrsz 1, kol 1 WST
pssw307@lab513-pc00:~/lab_pssw/lab5/alu/Synteza
Plik Edycja Widok Wyszukiwanie Terminal Pomoc
Module: alu
Operating conditions: uk65lscllmvbbr_120c25_tc
Interconnect mode: global
Area mode: physical library
Path 1: MET (10 ps) Path Delay Check
Startpoint: (F) A[0]
Endpoint: (R) P
Capture Launch
Path Delay:+ 1000 -
Drv Adjust:+ 0 0
Arrival:= 1000
Required Time:= 1000
Data Path:- 990
Slack:= 10
Exceptions/Constraints:
max_delay 1000 del_3
design:alu
alu
<Net>n_226
RTL
pssw307@lab513-pc00:~/lab_pssw...
Genus(TM) Synthesis Solution 22.1 ...
alu.v (~/lab_pssw/lab5/alu/RTL) - ge...


Programy Miejsca Terminal
czw, 09:40
Katalog domowy
Katalog domowy lab_pssw lab5 alu RTL
Ostatnio używane
Katalog domowy
Dokumenty
Muzyka
Obrazy
Pobrane
Wideo
Kosz
thinclient_drives
Inne położenia
alu.v
pssw307@lab513-pc00:~/lab_pssw/lab5/alu/Synteza
Plik Edycja Widok Wyszukiwanie Terminal Pomoc
Transform	Accepts	Rejects	Runtime (ms)
hlo_optimize_datapath_shifters	0	0	0.00
hlo_clip_mux_input	0	0	0.00
hlo_clip	0	0	0.00
hlo_cleanup	0	0	0.00
UM: flow.cputime flow.realtime timing.setup.tns timing.setup.wns snapshot
UM:* elaborate
design:alu
@genus:root: 4> check_design -unresolved
Check Design Report (c)
Unresolved References & Empty Modules
No unresolved references in design 'alu'
No empty modules in design 'alu'
Done Checking the design.
@genus:root: 5>
7
CENTOS
RTL
pssw307@lab513-pc00:~/lab_pssw...

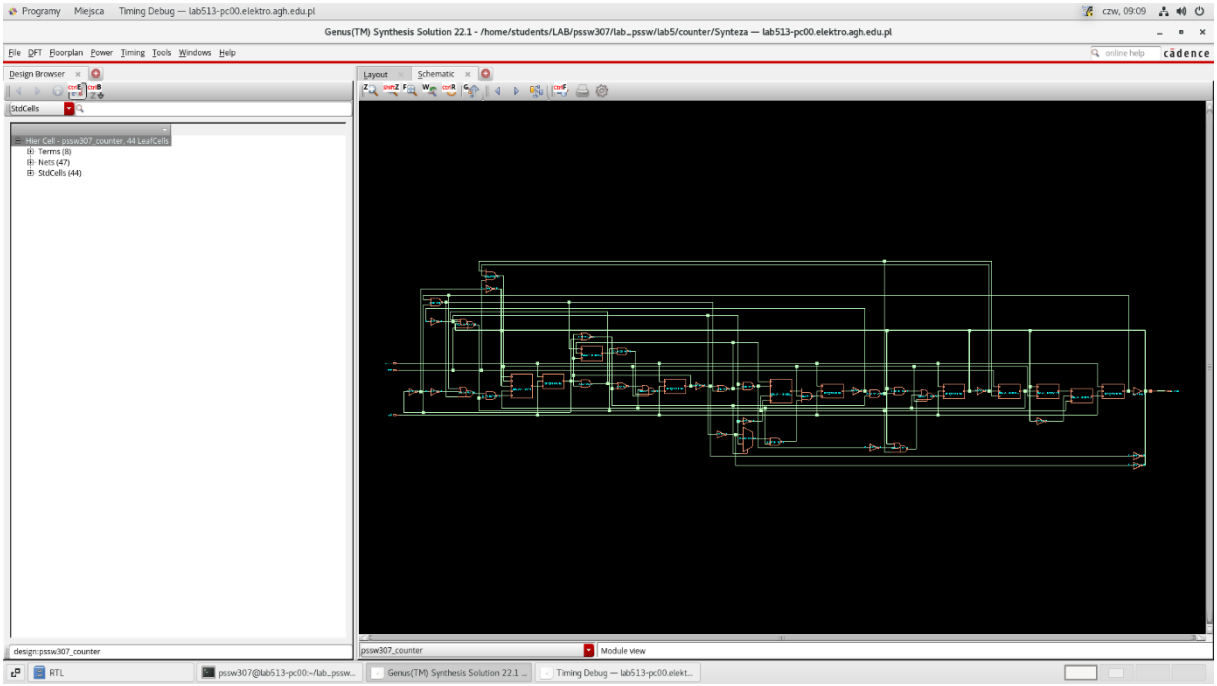
2. Licznik

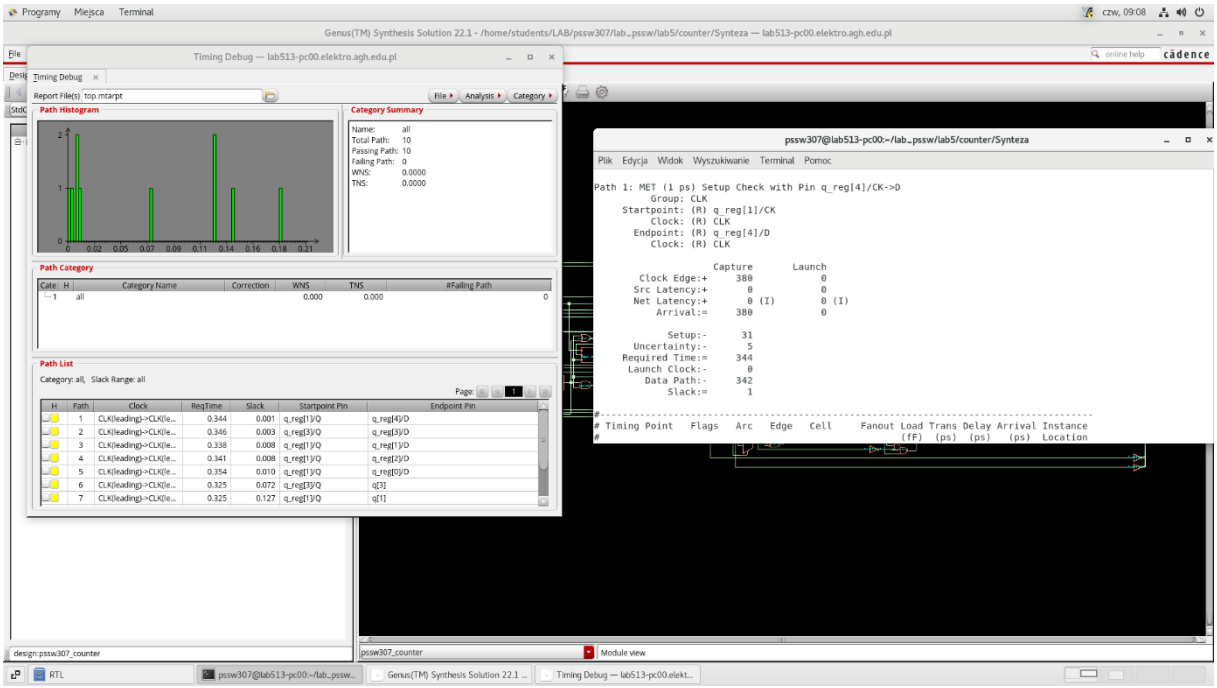
Min okres 0.38ns z zapasem 1ps. Max częstotliwość zegara to 2.63 GHz.

2.1. Przed syntezą



2.2. Po poprawnej optymalizacji i syntezie





2.3. Wyniki testów weryfikacyjnych

```
================================================================================
Mapped points: SYSTEM class
--------------------------------------------------------------------------------
Mapped points     PI     PO     DFF       Total
--------------------------------------------------------------------------------
Golden            3      5      5         13
--------------------------------------------------------------------------------
Revised           3      5      5         13
================================================================================
// Command: add compared point -all
// 10 compared points added to compare list
// Command: compare
================================================================================
Compared points      PO     DFF       Total
--------------------------------------------------------------------------------
Equivalent           5      5         10
================================================================================
// Command: report verification
================================================================================
                            Verification Report
--------------------------------------------------------------------------------
Category                                                                  Count
--------------------------------------------------------------------------------
1. Non-standard modeling options used:                                      0
--------------------------------------------------------------------------------
2. Incomplete verification:                                                 0
--------------------------------------------------------------------------------
3. User modification to design:                                             0
--------------------------------------------------------------------------------
4. Conformal Constraint Designer clock domain crossing checks recommended:  0
--------------------------------------------------------------------------------
5. Design ambiguity:                                                        0
--------------------------------------------------------------------------------
6. Compare Results:                                                       PASS
================================================================================
// Command: report verification > pssw307_counter_report.txt
// Command: exit
```

Pliki oraz raporty weryfikacyjne dla obu komponentów zamieszczone w archiwum *pssw307_files_and_reports_lab5* dołączonym wraz ze sprawozdaniem.